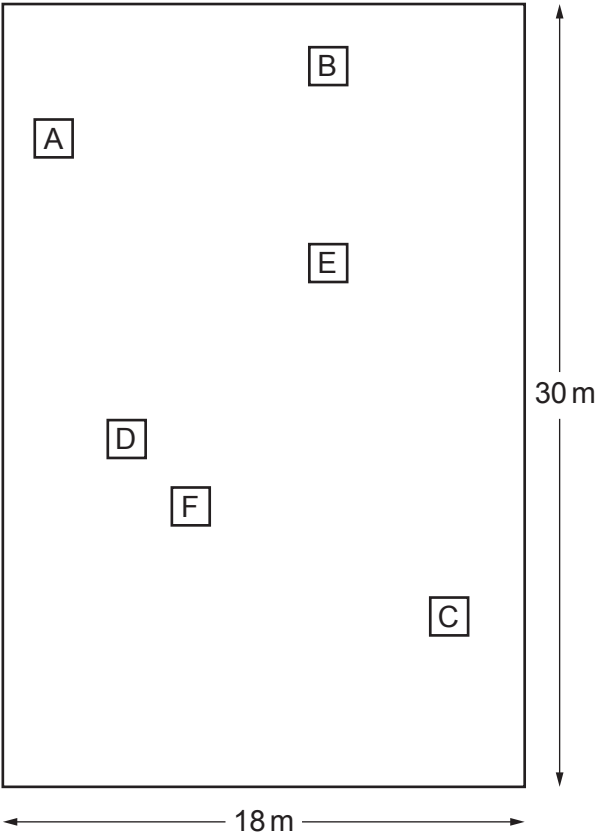


6. (a) Some students investigated the number of dandelion plants on a lawn. The diagram shows the lawn and the location of 6 quadrats (**A to F**) of area 1 m^2 which the students had placed at random on the lawn.



The students counted the number of dandelions in each quadrat and recorded their results in the table below.

Quadrat	Number of dandelions
A	6
B	3
C	2
D	7
E	4
F	5



- (i) Use the information to estimate the total number of dandelions on the lawn. [3]

estimated number of dandelions =

- (ii) The actual number of dandelions on the lawn is 1950.
Use the formula below to calculate the percentage error of your estimate above. [2]

$$\text{percentage error} = \frac{\text{estimated number of dandelions} - \text{actual number of dandelions}}{\text{actual number of dandelions}} \times 100$$

percentage error =

- (b) A species of dandelion has become resistant to one type of weedkiller. Explain, in terms of natural selection, how this has occurred. [4]

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